Estimating Vertical Chlorophyll Concentrations in Maize in Different Health States Using Hyperspectral LiDAR

Kaiyi Bi, Shunfu Xiao, Shuai Gao, Changsai Zhang, Ni Huang, and Zheng Niu

***Abstract*— The detection of vertical heterogeneity in vegetation has attracted an increasing attention as it has a great significance for precise agriculture. The hyperspectral light detection and ranging (LiDAR) (HSL) can obtain the spectral and spatial information simultaneously. However, its ability to monitor the vertical distribution of biochemical parameters in plants has not been fully explored. In this article, the applicability of empirical ratio and normalized spectral indices for HSL channels in chlorophyll (Chl) detection was investigated using three data sets: the PROSPECT-5 synthetic data set, the ANGERS public data set, and an HSL-measured data set. A linear regression model of the best performing index against measured Chl values was constructed so as to build 3-D Chl point clouds of maize. The performance of HSL in Chl detection at the upper and lower layers was also tested based on the selected spectral index. The result showed that the $CI_{red\ edge}$ index was most compatible with the HSL channels. The estimated Chl concentrations of the upper and lower layers showed the close relationships with HSL measurements ($R^2 = 0.73$ and 0.91, respectively). The vertical Chl profiles in maize were also presented, indicating that the HSL system has a strong ability to monitor the vertical distribution of maize Chl concentrations. This article provides a basis for the vertical detection of vegetation biochemical parameters directly from HSL measurements.**

***Index Terms*— Chlorophyll concentrations, health state, hyperspectral light detection and ranging (LiDAR) (HSL), maize, vertical distribution.**

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I. Introduction

CHLOROPHYLL (Chl) can be regarded as an indicator of the physiological stage and stress in plants. It is also essential to ecosystems as it influences plant energy and carbon [1]. Plants with a shortage of Chl show deficiencies in growth and carbon fixation. Remote sensing has become an efficient means of biochemical content detection and a standard for its nondestructive measurement [2]. The vertical distribution of vegetation properties has gained an increasing attention because plants generally have vertical heterogeneity, and this property affects the reflectance observed by remote sensing [2], [3]. Besides, the lower leaves of crops with insufficient nutrition turn yellow before higher ones [4] . Thus, assessing the vertical distribution of biochemical parameters is the key to assess the growth status of plants.

The passive remote sensing relying on illumination conditions has been widely used to characterize the biochemical properties due to its abundant spectral information [5]. Multiple- layer radiative transfer models have been developed to assess the importance of vertical heterogeneity in canopy reflectance [6], [7]. The vertical biochemical distribution has been investigated by top view observations; however, the upper part of the vegetation (the canopy) contributes more to the reflectance obtained by sensors than the lower parts [8]. On the contrary, multiangular observations improve the detection of plant vertical properties due to the variation in light scattering and shadowing depending on vegetation layers [9], [10]. However, the spectral information obtained by using multiple viewing angles is still a kind of mixed information from all vertical layers, and the effectiveness of using such data is limited due to its lack of actual spatial information [11]. Furthermore, in order to gain insight into the nonuniform distribution of leaf biochemical properties, the passive remotely sensed spectral information for each layer can be acquired by manually removing leaf layers from top to bottom, thus characterizing vegetation properties in the vertical direction [12], [13], but the knowledge it provided is difficult for practical applications [11]. Although the methods mentioned above has been used to investigate characteristics of different vertical layers, the spectral information provided by passive remote sensing can be confounded by other factors such as the background, canopy structure, and solar illumination [14], [15].

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As an active remote sensing technique, light detection and ranging (LiDAR) system has a small instantaneous field of view, making returned signals much purer than those of other techniques. Thus, it is possible for LiDAR point cloud to separate the returned signals of targets and soil [16]. LiDAR sensors are largely independent of illumination condition and can capture the backscattered intensity of targets at the "hotspot" [17]. It is widely known for extracting vegetation structural parameters such as tree height, leaf area index, and even plant phenotype characteristics due to its ability to penetrate the vegetation canopy and obtain accurate ranging information [18]–[21]. In recent years, LiDAR has gradually been used to detect various plant properties. Foliar Chl and nitrogen (N) show a strong relationship with the returned intensity of a green laser [15], [22], while vegetation biomass and moisture parameters can also be monitored with LiDAR systems [17], [23]. Furthermore, the spectral information provided by LiDAR makes the spectral three-dimensional (3-D) sensing possible, such as vegetation 3-D water mapping [14], [24]. However, single-wavelength LiDAR can be easily disturbed by confounding the factors such as detection distance, incidence angle, and mixed-edge returns. Multiwavelength LiDAR systems using separate LiDAR sensors may cause registration problems, and its available wavelength was restricted due to the limited manufacturing technology [25].

LiDAR with abundant spectral information could better explore absorption regions of plant reflectance spectrum, thus detecting biochemical properties more accurately. Hyperspectral LiDAR (HSL) using a supercontinuum laser (SC pulse) combines the advantages of hyperspectral remote sensing and traditional LiDAR. It has been successfully developed by several research groups [26]–[28] although a commercial HSL system has not appeared. The spectral and spatial information from more wavelength bands are available for any given 3-D point. Furthermore, the interactive process between the laser and target is not affected by illumination conditions and weather. The backscattered signal obtained by HSL can avoid many of the negative effects caused by multiple scattering and viewing angle [29], [30]. Most of the studies on HSL have been carried out on a two-dimensional (2-D) level due to the lack of experience and inefficiency in prototype HSL systems. Their results show that leaf biochemical parameters, such as N [31]–[33], carotenoids (Car) [34], and Chl [34], [35], can be successfully monitored using HSL systems. Besides, HSL also performs well in target classification as it has strong ranging ability, and the addition of spectral information greatly helps to improve the classification accuracy [36], [37]. The 3-D detection abilities of vegetation properties have also been assessed. Wallace *et al*. [38] showed that the NDVI profile measured using HSL has a good agreement with that measured by a spectrometer. Nevalainen *et al*. [35] monitored the 3-D Chl concentration of Scouts pine shoots using an HSL system and found that the Chl of each needle could be clearly depicted. Although the existing research on HSL systems is promising, to the best of our knowledge, no studies have explored the vertical distribution of crop biochemical parameters.

TABLE I
CENTER WAVELENGTHS (nm) OF THE 32 HSL BANDS

409	426	442	458
474	491	507	523
540	556	572	589
605	621	637	653
670	686	703	719
735	751	768	784
800	816	833	849
865	882	898	914

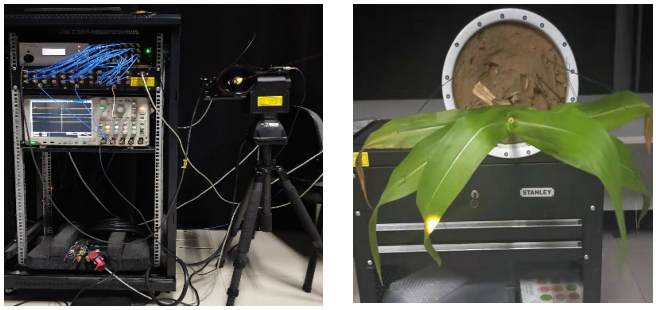


Fig. 1. (Left) Prototype HSL system. (Right) Way maize is placed during measuring.

The main goal of this article was to assess the capability of HSL to monitor the maize Chl concentrations at the upper and lower layers based on an analysis of the compatibility between HSL channels and traditional spectral indices. Maize, a food crop, is widely planted across the world and was used as a test species as the vertical heterogeneity within its canopy is distinct [8]. First, we tested the feasibility of selected ratio and normalized spectral indices in Chl estimation using three data sets: 1) PROSPECT-5 synthetic; 2) the ANGERS public data set; and 3) HSL measurements. The wavelength bands in the first two data sets were sampled with HSL channels. Then, using the upper and lower layers of maize with different health states, the relationships between the best-performing index and measured Chl concentrations were established; the vertical distribution of maize was also characterized.

II. MATERIALS AND METHODS

A. Hyperspectral LiDAR System

The HSL system designed by our team [26], [39] is a prototype that captures full-waveform LiDAR data from targets using 32 wavelength bands whose center wavelengths range from 409 to 914 nm, and the photograph of the HSL system is shown in Fig. 1. It can simultaneously acquire the spectral and ranging information. The system contains three main subsystems: a laser source, an optical receiver, and a signal acquisition system [26]. The light source is an SC laser (NKT SuperK). A telescope collects the scattered laser

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pulse returned from targets. Subsequently, the focus beam is almost equally split into 32 bands by an optical grating system, and the center wavelengths of the HSL channels are shown in Table I. However, due to the difference of the transmitted spectrum intensity and the photomultiplier tube array (PMT) sensitivity at various bands, HSL data have a lower signal-to-noise ratio at wavelength bands between 409 and 540 nm and 849 and 914 nm.

B. PROSPECT-5 Model

Although the experimental targets used in this article were maize plants, a leaf radiative transfer model rather than a canopy model was used due to the small target instantaneous field of view of HSL. The PROSPECT-5 model is a leaf model that can accurately describe the directional-hemispherical reflectance and transmittance of target leaves from 400 to 2500 nm. The PROSPECT model is a plate model that describes the leaf as a stack of N homogeneous, absorbing the elementary layers [40]. Its input parameters include the leaf structure parameter N, Chl (μg/cm^2), Car (μg/cm^2), equivalent water thickness (EWT, cm), and dry matter content (LMA, g/cm^2). By running the PROSPECT-5 model in the direct mode and assigning these leaf parameters, corresponding leaf reflectance and transmittance values can be simulated based on the optical constants consisting of refractive indices and specific absorption coefficients.

C. Data Collection

1) Synthetic and Public Data Sets: Narrowband indices are mainly designed based on a traditional passive spectrum with 1-nm intervals, while the intervals of the HSL data set are 16 nm or 17 nm. Besides, the calculations of spectral indices are only based on the closest channel instead of the required channel, which may cause error and uncertainty. For example, in some spectral regions such as red-edge region, whose reflectance change is abrupt, even a small change in a channel can result in great uncertainty in the results. Thus, it is necessary to explore whether these selected spectral indices are suitable for use with HSL data to estimate the vegetation Chl parameters at the leaf level and to determine which index is most compatible.

In order to find out whether spectral indices constituted of HSL bands are effective for monitoring vegetation leaf biochemical parameters at the leaf level in theory, a synthetic data set ($n = 1000$), obtained by running PROSPECT-5 in direct mode, and a public data set, ANGERS ($n = 276$), were used. The reflectance of the synthetic and public data sets was resampled in accordance with the spectral width of the sensor, and then, both the data sets contained 32 bands—the same as the HSL data.

The input parameters of the PROSPECT-5 model were assumed to have a Gaussian distribution, as shown in Table II. The EWT values were kept constant as 0.003 due to them being insensitive in the range of 409–914 nm. The structure parameter N was not fixed because its estimation is the basis for the accurate estimation of the other parameters. The other parameters were set based on the studies of Sun *et al*. [25] and Féret *et al*. [40]. Compared with the PROSPECT-5 synthetic data set, the public ANGERS 2003 data set contained more noise. This data set, containing 43 different species, was measured by Jacquemoud on June 2003. The reflectance and transmittance spectra ranging from 400 to 2500 nm in 1-nm steps were acquired for each leaf, and then, the leaf Chl, Car, EWT, and LMA were measured.

TABLE II
DISTRIBUTION CHARACTERISTICS OF LEAF PARAMETERS FOR SYNTHETIC DATA SETS

Unit	N -	Chl μg/cm^2	Car μg/cm^2	LMA g/cm^2
Mean	1.60	32.81	8.51	0.0077
Standard deviation	0.30	18.87	3.92	0.0035
Minimum	1	0	0	0.0008
Maximum	3.5	100	30	0.0331

2) HSL Data Set: The data used in this article were collected during the 2018 growing season. About 20 pots of maize were planted at the HuaiLai Remote Sensing Test Station, Chinese Academy of Sciences, Huailai County, Hebei Province (40.35° N, 115.78° E). One maize plant was planted per flowerpot. There were two groups of maize used in this article. The first contained ten pots of healthy maize with a normal N fertilization amount, whereas the other group of ten pots was set to grow with insufficient N fertilization. Given that these potted corn plants would be experimented indoors, their growing period only lasted about 2 months so that their height (average height was about 100 cm) was suitable for laboratory measurements.

The HSL measurements were conducted in a dark experimental condition. Each pot of maize was placed horizontally at a distance of ~5 m from the HSL system because it is not easy for the prototype HSL system to emit a laser pulse vertically (Fig. 1). Considering the efficiency of the prototype system, each point was only scanned once. Full-waveform data at each wavelength band were first extracted from the return of each target point; then, the echo position of full waveform was better identified by applying the Gaussian fitness. Thus, the ranging and spectral information could be extracted based on the returned time and intensity of the echo position. After being separated from the background, each maize point cloud was clipped into two equal layers according to the height of the maize stem, and then, the intensity values of the point cloud at each layer were averaged.

After the HSL measurement, each pot of maize was placed vertically on the ground and clipped manually into two equal layers consistent with the segmented point cloud. Subsequently, for each level, maize leaves were packed and sent to the laboratory for Chl determination, which was based on their absorption properties with 95% ethanol.

D. Selected Vegetation Index

The returned laser intensity at each wavelength was affected by both the leaf edges and the incidence angle. When the laser beam partially occupies a leaf edge, the returned laser beam

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TABLE III
SPECTRAL INDICES USED IN THIS ARTICLE

Spectral index	Formula	References
SR (simple ratio)	R_{810}/R_{560}	[41]
SR2	R_{750}/R_{710}	[42]
MSR (modified simple ratio)	$\frac{(R_{800}/R_{670})-1}{\sqrt{(R_{800}/R_{670})+1}}$	[43]
$CI_{red\ edge}$ (red-edge chlorophyll index)	$(R_{780}/R_{710}) - 1$	[44, 45]
CI_{green} (green chlorophyll index)	$(R_{780}/R_{550}) - 1$	[45]
NDRE (normalized difference red edge)	$(R_{790} - R_{720})/(R_{790} + R_{720})$	[46]
NDVI [705, 750] (normalized difference vegetation index [705, 750])	$(R_{750} - R_{705})/(R_{750} + R_{705})$	[47]
NDVI[670, 800] (normalized difference vegetation index [670, 800])	$(R_{800} - R_{670})/(R_{800} + R_{670})$	[48]
MTCI (MERIS terrestrial chlorophyll index)	$(R_{754} - R_{709})/(R_{709} - R_{681})$	[49]
PRI (photochemical reflectance index)	$(R_{531} - R_{570})/(R_{570} + R_{531})$	[50]

also contains a reflectance signal from background objects, causing "mixed edge returns" [16]. The incidence angle is another influence and is defined as the angle between the laser beam and the normal vector of the leaf. An incidence angle that deviates from the perpendicular orientation can result in energy dispersion. As the leaf angle increases, the laser intensity decreases accordingly [24].

The radiometric correction is necessary with single wavelength instruments to eliminate these confounding factors [14], while dual- or multi-wavelength laser systems can potentially reduce these factors by using a ratio index, as different wavelengths are affected by the same factors along the optical path [17], [35]. Thus, several empirical vegetation indices that are ratio related were selected for use in this article (Table III). For the synthetic and public data sets, as well as HSL-collected data set, the channel closest to the wavelength required by the index was utilized.

E. Sensitivity Analysis

The sensitivity analysis can identify the contribution of each input parameter to the output variables [51]. Running the PROSPECT-5 data set in the forward model and assigning the variables as the input parameters *N*, Chl, Car, EWT, and LMA (Table II), a list of wavelength ranges corresponding to the HSL channels (Table I) can be derived. Then, the corresponding sensitivity analysis results are produced based on the derived data set, showing the relative importance of each parameter to the reflectance in a certain HSL channel. The process of the sensitivity analysis was conducted in MATLAB.

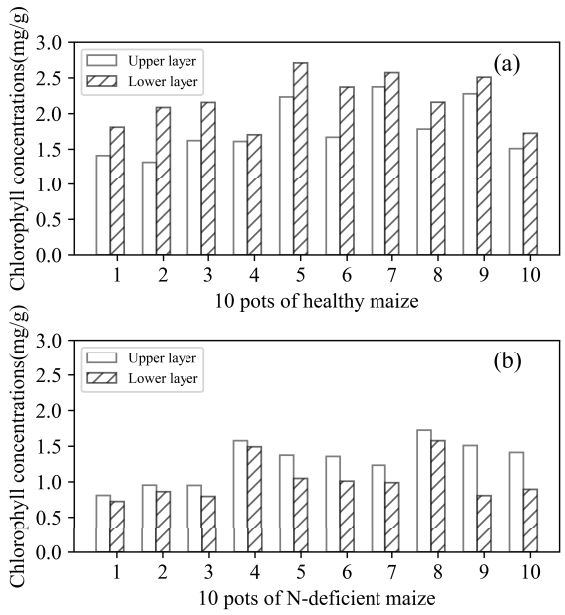


Fig. 2. Measured Chl concentrations in the upper and lower layers of (a) healthy maize and (b) N-deficient maize.

III. RESULTS

A. Chl Vertical Distribution Within Maize

To investigate whether HSL has a vertical detection ability, it is necessary to know the Chl distribution within the maize used in this article. The Chl concentrations of both treatment groups are shown in Fig. 2. The healthy and N-deficient maize presented the significant differences in Chl concentration, with healthy maize having higher Chl concentrations than the N-deficient group. The range of measured Chl values of the upper and lower layers was 0.72–1.72 and 1.32–2.71 mg/g for N-deficient and healthy maize, respectively. It is also worth noting that the upper layers of healthy maize showed higher Chl concentrations than the lower layers, which was opposite to the case in N-deficient maize (Fig. 2).

B. Performance of Spectral Indices Based on HSL Channels

A PROSPECT-5 synthetic data set was first used. As shown in Fig. 3, HSL channels cover a wide range of the spectrum. Several channels are sensitive to both Chl parameters and leaf structure parameters, which are essential for Chl estimation. The performances of all the tested spectral indices are shown in Table IV. On the whole, several spectral indices performed well (in terms of R^2) to inverse the leaf biochemical parameters, indicating that the spectral information of the HSL channels is theoretically sufficient to estimate Chl through ratio- or normalized-based spectral indices. The closest relationship was achieved using NDRE [$R^2 = 0.97$, Fig. 4(a)], followed by $CI_{red\ edge}$, CI_{green}, and SR2 ($R^2 = 0.96$). Due to the fact that the PROSPECT-5 synthetic data set did not

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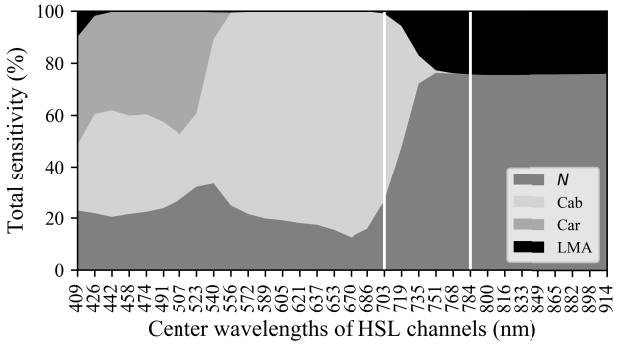


Fig. 3. Global sensitivity analysis of the input parameters for PROSPECT-5 reflectance corresponding with HSL channels. White lines indicate HSL channels centered at 703 and 784 nm.

TABLE IV

COEFFICIENTS OF DETERMINATION (R^2) OF THE LINEAR RELATIONSHIP BETWEEN THE TESTED VEGETATION INDICES BASED ON HSL CHANNELS AND CHL. THE HIGHEST R^2 VALUES OF THREE DATA SETS ARE SHOWN IN BOLD

Dataset	SR1	SR2	MSR	$CI_{red\ edge}$	CI_{green}
PROSPECT	0.95	0.96	0.74	0.96	0.96
ANGERS	0.93	0.93	0.37	**0.95**	0.92
HSL	0.20	0.79	0.50	**0.84**	0.76
Dataset	**NDRE**	**NDVI [705, 750]**	**NDVI [670, 800]**	**MTCI**	**PRI**
PROSPECT	**0.97**	0.92	0.50	0.55	0.48
ANGERS	0.94	0.75	0.21	0.35	0.40
HSL	0.73	0.79	0.50	0.78	0.17

include various types of noise, the ANGERS public data set was used for further investigation. The relationship between the spectral indices based on HSL channels and biochemical concentration is shown in Table IV. The best performance was achieved by $CI_{red\ edge}$ [$R^2 = 0.95$, Fig. 4(b)], followed by NDRE ($R^2 = 0.94$). From the results of these two data sets, it seems that HSL channel-based $CI_{red\ edge}$ and NDRE are more effective for Chl estimation.

The PROSPECT-5 data set contains a wide variety of input parameters, and the ANGERS data set includes 276 leaves of different species. Subsequently, the inversion results with the HSL-measured data set are also illustrated.

The ability of the selected spectral indices to estimate leaf biochemical parameters based on the HSL-measured data set is also illustrated (Table IV). The best performance was achieved using $CI_{red\ edge}$, which is similar to the ANGERS public data set. Some indices performed poorly possibly because they are incompatible with HSL channels. A scatterplot of the best performing spectral index is shown in Fig. 5. As can been seen, $CI_{red\ edge}$ is positively related to Chl ($R^2 = 0.84$ and RMSE = 0.13).

Compared with the public and synthetic data sets, some spectral indices showed different abilities in estimating leaf Chl concentrations, partly due to the lower signal-to-noise ratio of some of the HSL channels of the prototype system

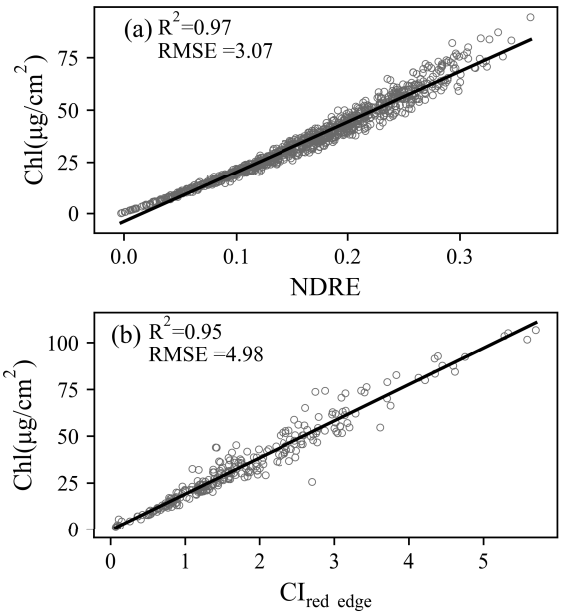


Fig. 4. Relationship between the Chl concentrations and the best performed spectral index for (a) PROSPECT-5 synthetic data set ($n = 1000$) and (b) ANGERS public data set ($n = 276$).

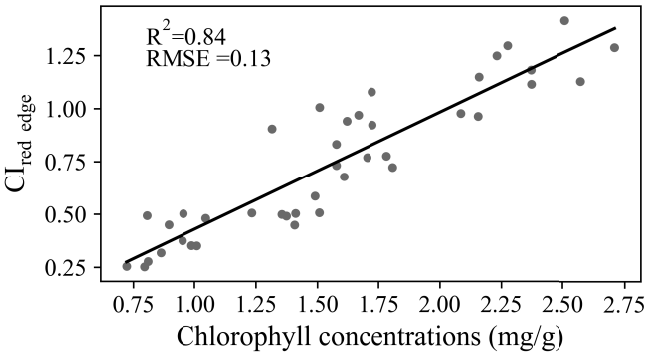


Fig. 5. Relationship between Chl concentrations and $CI_{red\ edge}$ based on the HSL data set for all pots of maize. Both the upper and lower layers were considered, and point cloud values at each layer were averaged.

and differences in the experimental targets. Besides, the HSL-acquired data set contains the backscattered intensity instead of hemispherical reflectance. Regardless, the capability of the spectral indices based on HSL channels has been verified. It is worth mentioning that $CI_{red\ edge}$ based on HSL channels was confirmed to be the most effective of the three data sets in monitoring leaf biochemical contents. The HSL channel used in this spectral index was 703 nm, which is sensitive to Chl content, and 784 nm, which has a high correlation with the leaf structure parameter N (Fig. 3). Thus, in the next section, $CI_{red\ edge}$ is used to invert the vertical biochemical concentrations.

C. Vertical Biochemical Content Estimation in Maize

The relationship between $CI_{red\ edge}$ and Chl concentrations of upper and lower maize plant layers is presented in Fig. 6. The high R^2 (0.73 and 0.91) and low RMSE (0.15 and 0.11)

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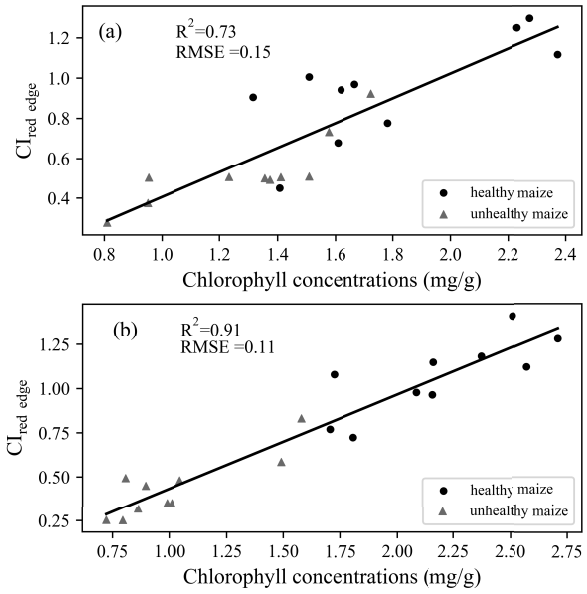


Fig. 6. Correlation between Chl and $CI_{red\ edge}$ calculated based on the HSL data set for the upper and lower layers of maize. Black dots indicate healthy maize and gray triangles indicate N-deficient maize. All points (black and gray) were used to fit the linear regression model.

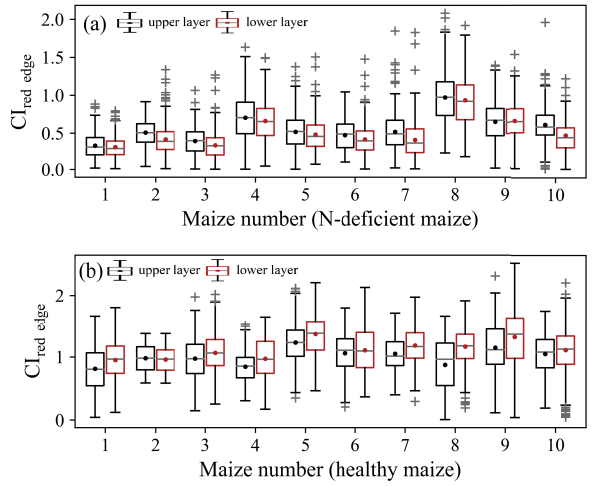


Fig. 7. Distribution of $CI_{red\ edge}$ values of the upper and lower layers for (a) N-deficient and (b) healthy maize. The box boundaries represent the 25th and 75th percentiles, black and red circles represent means, and gray lines represent medians. Gray crosses are outliers, and the whiskers represent the extreme values of the point cloud.

indicate high accuracy, meaning that $CI_{red\ edge}$ can predict leaf Chl concentrations at different heights of maize. The lower layers showed a good performance due to the high penetration ability of the HSL. Besides, the range of $CI_{red\ edge}$ values in healthy maize was higher than that in N-deficient maize for both the upper and lower layers. Hence, the health status of maize can be more easily distinguished using $CI_{red\ edge}$ values from the lower layers.

To analyze the uncertainty in the estimation, the distributions of $CI_{red\ edge}$ values within healthy and N-deficient maize were described (Fig. 7). There were some outliers existing in

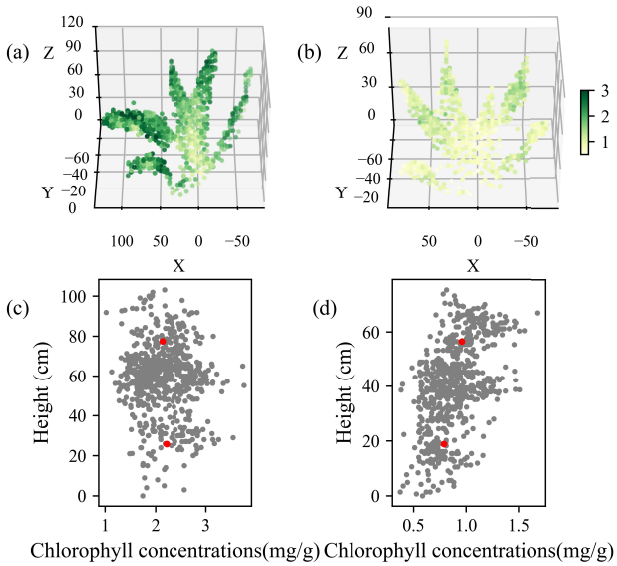


Fig. 8. Examples of 3-D Chl concentration(mg/g) estimates in (a) healthy maize, (b) N-deficient maize, and (c) and (d) their corresponding Chl distributions excluding points on the stem (red points indicate the means of the upper and lower layers). The Chl content at each point was estimated using the $CI_{red\ edge}$ linear regression model (see Fig. 5): Chl concentration = $CI_{red\ edge} \times 1.51 + 0.44$. X and Y represent the horizontal plane and Z represents the vertical direction (in cm).

each point cloud. Besides, for each N-deficient maize plant, the mean $CI_{red\ edge}$ value of the upper layer was higher than that of the lower layer [Fig. 7(a)], which was opposite to the case for healthy maize [Fig. 7(b)]. This trend is consistent with the measured Chl concentrations (Fig. 2), meaning that the mean value of $CI_{red\ edge}$ within each point cloud can efficiently represent the properties of the upper and lower layers of maize.

In order to visually show the 3-D distribution of Chl concentrations, the point clouds of two maize plants are shown in Fig. 8(a) and (b) to give a demonstration. As can be seen, the variation in Chl concentration in each layer and even each 3-D point can be monitored. Furthermore, the points at leaf edges or curved places appear normal, as is the case for other points, which indicates the efficiency of the selected spectral index in reducing the effects of leaf edge and incidence angle. The center parts of healthy maize plants, which are the stems, showed the lower Chl concentrations than other parts. However, in this article, only leaves were considered. Compared with the healthy maize, the N-deficient one showed the lower Chl concentrations. The corresponding vertical Chl distributions of the two plants are shown in Fig. 8(c) and (d). The Chl concentration in the N-deficient maize increased with the increase of height, which was opposite to the healthy one. Although the heights of experimental maize plants were short, the Chl concentration acted as a function of maize height, further indicating the capability of HSL to monitor the vertical distribution of biochemical parameters in vegetation.

IV. Discussion

Three data sets were utilized in this article to investigate the applicability of ratio and normalized spectral indices for HSL channels. Besides HSL measurements, the PROSPECT-5 synthetic and ANGERS public data sets were used.

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The compatibility between selected spectral indices and channel bands was tested although there were some differences between the passive and active data sets because of their distinct detection mechanisms. The unit used in the PROSPECT-5 and ANGERS data sets is μg/cm^2, whereas that in the HSL measured data set is mg/g; however, these two quantities have a positive relationship and can be computed using 1/LMA [51], [52].

At present, the multichannel data acquisition resulted in the sacrifice of the measuring time [26], and thus, only a small data set was tested in this experiment. A close relationship based on a linear regression model was observed between chlorophyll concentrations and spectral indices (Fig. 5), and the variation of chlorophyll concentration at any height (z value) could be observed based on the 3-D point cloud (Fig. 8). It means that the effectiveness of HSL in plant vertical Chl detection was justified both statistically and visually. Although the band interval of HSL data was 16 or 17 nm, the spectral information of PROSPECT-5 synthetic data sampled in accordance with the HSL sensor was still sensitive to biochemical parameters (Fig. 3), indicating that the band interval of HSL data is sufficient to retrieve the biochemical parameters.

In terms of inversion methods, the empirical ratio and normalized spectral indices which aimed to eliminate the confounding factors of leaf edges and incidence angle were used. These indices can normally achieve good results for HSL-measured data [34], [53]. The best performing index ($CI_{red\ edge}$) has also been demonstrated by other studies to have a strong ability for vertical biochemical parameter estimation [8]. Inversion methods employing more spectral information have been tested by several HSL-based studies [30], [51], [54], but their experimental targets were limited in 2-D level.

Some limitations were also noted in this experiment. Maize plants investigated in this article were shorter and smaller than natural maize due to their limited growing conditions; hence, only two vertical layers were considered. Although Chl concentrations varied with height differently in healthy and N-deficient maize [Fig. 8(c) and (d)], the short heights prevented the vertical profiles from showing a "bell-shaped" appearance, which normally occurs with maize [55], [56].

Few outliers existed in each maize point cloud that contained hundreds of the 3-D points (Fig. 7). Besides the system error, the full-waveform decomposition method which is the Gaussian decomposition in this article may also affect the quality of maize point clouds. Hence, the improvement of the accuracy of the extracted LiDAR information needs advanced calibration and decomposition methods [14], [57]. What is more, plant structural and biochemical parameters could complement for each other and improve the understanding and modeling of plant ecophysiological processes [15], [23], while structural parameters of maize plants could not be accurately estimated in this experiment. This was mainly caused by the relatively sparse point cloud density and the horizontal placement of maize plants during measurement.

With the development of hardware techniques, HSL systems would be more flexible and effective. The HSL systems were designed with wide spectral ranges and detection channels. It means that a variety of extraction methods, which explore more absorption regions of plant reflectance spectrum thus detecting biochemical parameters, could be utilized in the future based on advanced calibration and decomposition methods and higher signal-to-noise ratio of HSL bands. Based on this article, future experiments can be gradually extended to the field with large data sets, thus monitoring the vertical biochemical profile of extensive crop types at various growing stages or under different conditions. Obtaining spatial and abundant spectral information simultaneously, HSL also holds promise to extract the crop structural parameters along with biochemical components, thus better characterizing crop properties.

V. Conclusion

This article investigated the performance of spectral indices based on HSL channels and assessed the ability of the HSL system to monitor vertical Chl distributions in maize. In order to evaluate the ability of the selected spectral indices, three data sets were used. The best performing spectral index was used to invert the Chl concentrations in the upper and lower layers of maize in two health states. The results show that the $CI_{red\ edge}$ index was most compatible with HSL channels, and the HSL system has a high ability to evaluate the Chl concentrations at different heights based on its precise ranging information.

The research provides an insight into the ability of the HSL system in estimating Chl parameters at different heights, as well as monitoring the health status of plants at an early growth stage. On the contrary, conventional passive remote sensing is limited due to reduced dimensionality. The application of the HSL system is not limited to the estimation of Chl and could be used to estimate other biochemical concentrations because of its abundant spectral information. With the simultaneous acquisition of spectral and spatial information, HSL systems have a great potential to accurately estimate the biochemical parameters at any height, thus helping to improve the estimates of carbon stocks and biomass for ecosystem management.

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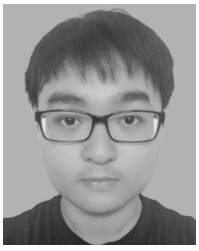
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